

* Logic

• Statement

A **statement** is a **declarative sentence** that is either **true** or **false**, but not both. $\rightarrow$ 陳述句

True and **false** are truth value (真值).

eg: 1. X is good,

Not statement

2. $2+3=5$

True statement

3. $2 \in \mathbb{N}$

True statement

4. $\pi \in \mathbb{Q}$

False statement

5. $1+1=0$

False statement

6. $\sin X = 1$

Not statement (Open Sentence)

• Logic Connectives and Truth Table

1. Not, negation ($\sim, \neg, !$)

P : True, $\sim P$: False

P	$\sim P$
T	F
F	T

truth table

eg. ① $P: 1+1=2$, $\sim P: 1+1 \neq 2$

② P : Today is Monday

$\sim P$: Today is not Monday

Other notations: $\neg P$, $!P$, P' , $\bar{P}$

2. And ($\wedge$)

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

eg. I like play games and balls.
 $\underbrace{\text{games}}_P \wedge \underbrace{\text{balls}}_Q$

3. Or ($\vee$)

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

eg. Tom's PC runs Linux or MacOS.
 $\underbrace{\text{Linux}}_P \vee \underbrace{\text{MacOS}}_Q$

eg. $(\neg P \vee Q) \wedge (P \wedge \neg Q)$

P	Q	$\neg P$	$\neg Q$	$\neg P \vee Q$	$P \wedge \neg Q$	$(\neg P \vee Q) \wedge (P \wedge \neg Q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	T	F	F
F	F	T	T	T	F	F

• Conditional Statements

1. If, then ($\Rightarrow$): imply → sufficient

If P, then Q. : $P \Rightarrow Q$ → necessary

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Remark:

(1) "If P, then Q" is equivalent to

① "Q if P"

② "Q, provided that P"

(2) $P \Rightarrow Q$ and $\sim P \vee Q$ have two same truth values.

They are logically equivalent.

2. P if and only if Q ($P \Leftrightarrow Q$)

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$	$P \Leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Remark: $P \Leftrightarrow Q$ and $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ are logically equivalent.

- Logical Equivalence

Def (Tautology) $\rightarrow$ 永真式

A **tautology** is a statement that is always true.

eg: 1. $P \vee (\sim P)$

P	$\sim P$	$P \vee (\sim P)$
T	F	T
F	T	T

$\rightarrow$ Tautology

2. $(\sim P \vee Q) \vee (P \wedge \sim Q)$

3. $((P \Rightarrow Q) \wedge Q) \Rightarrow Q$

4. $(P \Rightarrow Q) \vee (Q \Rightarrow P)$

prove with truth table

Def (Logical Equivalence)

Let S_1, S_2 are statements. S_1 and S_2 are **logically equivalent** if and only if $S_1 \Leftrightarrow S_2$ is tautology. We denote

$$S_1 \equiv S_2$$

as $S_1 \Leftrightarrow S_2$ is tautology.

Remark:

In fact, S_1 and S_2 are logically equivalent if they have the same truth value in truth table.

eg. Some important tautologies.

$$(1) \neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$$

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg(P \vee Q)$	$(\neg P) \wedge (\neg Q)$	$\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q)$
T	T	F	F	T	F	F	T
T	F	F	T	T	F	F	T
F	T	T	F	T	F	F	T
F	F	T	T	F	T	T	T

↓
tautology

$$(2) \neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$$

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$	$\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$
T	T	F	F	T	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	T	T

Note: (1) and (2) we call it is "De Morgan's Law".

Remark:

(1) Contrapositive: $\neg Q \Rightarrow \neg P$ is **contrapositive** if $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are logically equivalent, we can write $P \Rightarrow Q \equiv (\neg Q) \Rightarrow (\neg P)$

(2) Contradiction: A statement is **contradiction** if it is always false.
(矛盾式)

Here are some logical equivalences.

1. $P \Rightarrow Q \equiv \sim P \vee Q$

$$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

2. Contrapositive Law

$$P \Rightarrow Q \equiv (\sim Q) \Rightarrow (\sim P)$$

3. De Morgan's Laws

$$\sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$$

$$\sim(P \vee Q) \equiv (\sim P) \wedge (\sim Q)$$

4. Commutative Laws

$$P \wedge Q \equiv Q \wedge P \quad P \Leftrightarrow Q \equiv Q \Leftrightarrow P$$

$$P \vee Q \equiv Q \vee P$$

5. Distributive Laws / Absorption Laws

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

$$P \wedge (P \vee Q) \equiv P$$

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

$$P \vee (P \wedge Q) \equiv P$$

6. Associative Laws

$$P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$$

$$P \vee (Q \vee R) \equiv (P \vee Q) \vee R$$

7. $\sim(\sim P) \equiv P$ (Law of double negation)

• Negating Statements

Def (Quantifiers)

1. $\forall$: "For all", "^(each)for any", "for every". 3. $\exists!$: "there is an unique"
2. $\exists$: "There exists" or "There is a".

eg. "For every $n \in \mathbb{N}$, $2n$ is even."

$$\Leftrightarrow \forall n \in \mathbb{N}, 2n \text{ is even.}$$

eg. "There is an integer x is greater than 0."

$$\Leftrightarrow \exists x > 0.$$

Recall: The negation of P and DeMorgan's Law.

Let all sentences as R , and R has P and Q .

eg. $R \equiv P \Rightarrow Q$

Then $\sim R \equiv \sim(P \Rightarrow Q)$

$$\equiv \sim(\sim P \vee Q)$$

$$\equiv P \wedge (\sim Q)$$

eg. Negate $(x \text{ is odd}) \wedge (y \text{ is odd})$.

$$\sim[(x \text{ is odd}) \wedge (y \text{ is odd})] \equiv (x \text{ is even}) \vee \sim(y \text{ is odd})$$

$$\equiv (x \text{ is even}) \vee \sim(y \text{ is odd})$$

$$\equiv (x \text{ is even}) \vee (y \text{ is even})$$

$$\text{eg. 11) } \sim (\forall x \in X, P) \equiv \exists x \in X, \sim P$$

$$\text{12) } \sim (\exists x \in X, P) \equiv \forall x \in X, \sim P$$

$$\text{eg. } \sim (\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon)$$

$$\equiv \exists \varepsilon > 0, \forall \delta > 0 \text{ at } 0 < |x - a| < \delta \text{ but } |f(x) - L| \geq \varepsilon.$$

$$\left(\begin{aligned} 0 < |x - a| < \delta &\Rightarrow |f(x) - L| < \varepsilon \\ &\equiv \sim (0 < |x - a| < \delta) \vee (|f(x) - L| < \varepsilon) \\ &\Rightarrow \sim [\sim (0 < |x - a| < \delta) \vee |f(x) - L| < \varepsilon] \\ &\equiv (0 < |x - a| < \delta) \wedge (|f(x) - L| \geq \varepsilon) \end{aligned} \right)$$